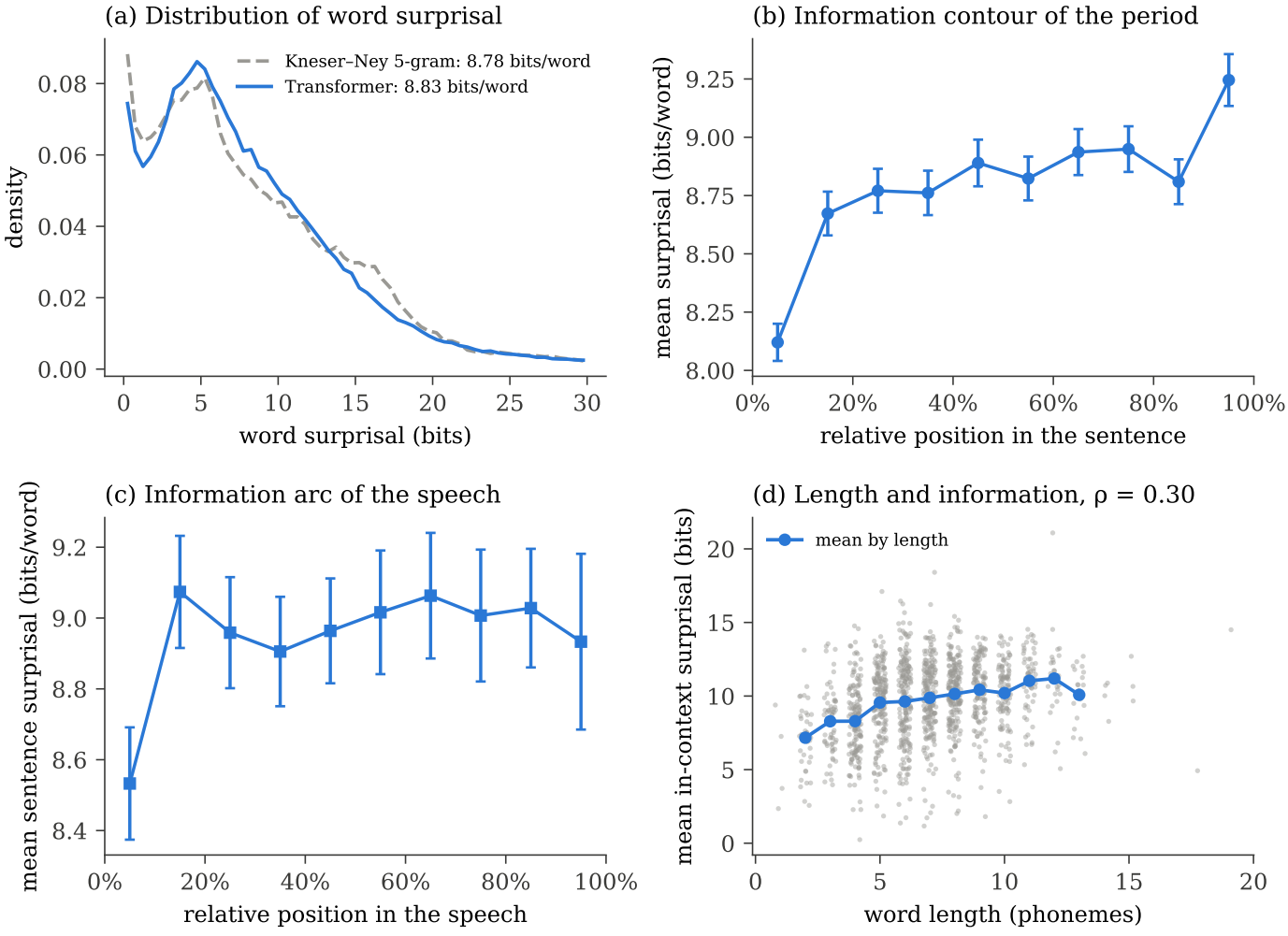
(a) Distribution of word surprisal
Kneser–Ney 5-gram: 8.78 bits/word
Transformer: 8.83 bits/word
density
0.00
0.02
0.04
0.06
0.08
0
5
10
15
20
25
30
word surprisal (bits)
(b) Information contour of the period
mean surprisal (bits/word)
8.00
8.25
8.50
8.75
9.00
9.25
0%
20%
40%
60%
80%
100%
relative position in the sentence
(c) Information arc of the speech
mean sentence surprisal (bits/word)
8.4
8.6
8.8
9.0
9.2
0%
20%
40%
60%
80%
100%
relative position in the speech
(d) Length and information, ρ = 0.30
mean by length
mean in-context surprisal (bits)
0
5
10
15
20
0
5
10
15
20
word length (phonemes)